

Gaussian Beams Beyond the Paraxial Approximation

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(Dated: September 25, 2026)

Beyond the paraxial limit the name “Gaussian beam” does not define a unique field. One must say where the Gaussian is: in the angular spectrum, or in the pupil of a lens. We write six such beams as exact solutions of Maxwell’s equations, each fixed by an envelope $a(\theta)$ and a polarization rule $c(\theta)$. They include the beams of COMSOL and of the Optical Tweezers Toolbox.

I. PARAXIAL GAUSSIAN BEAM

The time dependence is $e^{-i\omega t}$, and the units are SI. The medium is homogeneous and lossless, with the absolute permittivity ε , the absolute permeability μ , the index $n = c\sqrt{\varepsilon\mu}$ and the wavenumber $k = \omega\sqrt{\varepsilon\mu} = 2\pi n/\lambda$. The position is $\mathbf{r} = (\rho, \varphi, z)$ in cylindrical coordinates.

The paraxial Gaussian beam reads [1]

$$\sqrt{\varepsilon}\mathbf{E} = \mathcal{A}_0 \mathbf{u} \frac{w_0}{w(z)} e^{-\frac{\rho^2}{w^2(z)}} e^{i[kz - \eta(z) + R^{-1}(z)k\rho^2/2]}, \quad (1)$$

$$\sqrt{\mu}\mathbf{H} = \hat{\mathbf{z}} \times \sqrt{\varepsilon}\mathbf{E},$$

where $\mathcal{A}_0/\sqrt{\varepsilon}$ is the electric field amplitude, $\mathbf{u}$ is the unit complex polarization vector, $w(z) = w_0\sqrt{1 + z^2/z_0^2}$ is the beam radius, $R^{-1}(z) = z/(z^2 + z_0^2)$ is the inverse wavefront radius, $\eta(z) = \arctan(z/z_0)$ is the Gouy phase, and $z_0 = kw_0^2/2$ is the Rayleigh range. We use the circular polarization vectors

$$\hat{\mathbf{e}}_\sigma = (\hat{\mathbf{x}} + i\sigma\hat{\mathbf{y}})/\sqrt{2}, \quad (2)$$

with the handedness $\sigma = \pm 1$.

Equation (1) solves the paraxial wave equation, not Maxwell’s equations, and it needs $kw_0 \gg 2$. Beyond this limit no closed-form Gaussian beam exists, and the name “Gaussian beam” no longer defines a unique field. One must say where the Gaussian is. It can be in the angular spectrum, in the pupil of an aplanatic lens, or in the pupil of a thin lens. The code for all beams below is in Ref. [2].

II. NON-PARAXIAL GAUSSIAN BEAMS

A monochromatic beam radiated by sources at $z \rightarrow -\infty$ is a sum of propagating plane waves [3, 4],

$$\mathbf{E}(\mathbf{r}) = \int_{\theta < \theta_{\max}} d\Omega \mathbf{E}_\infty(\theta, \phi) e^{i\mathbf{k} \cdot \mathbf{r}}, \quad (3)$$

with $\mathbf{k} = k\hat{\mathbf{k}}$, $\hat{\mathbf{k}} = \cos\phi \sin\theta \hat{\mathbf{x}} + \sin\phi \sin\theta \hat{\mathbf{y}} + \cos\theta \hat{\mathbf{z}}$ (Fig. 1A), $d\Omega = \sin\theta d\theta d\phi$ and $\hat{\mathbf{k}} \cdot \mathbf{E}_\infty = 0$. The limit is $\theta_{\max} = \pi/2$, unless a stop sets $\theta_{\max} = \arcsin(\text{NA}/n)$. Each plane wave solves Maxwell’s equations, so Eq. (3) is exact for every transverse $\mathbf{E}_\infty$. The magnetic field is Eq. (3) with $\mathbf{E}_\infty$ replaced by $\mathbf{H}_\infty$, where $\sqrt{\mu}\mathbf{H}_\infty = \hat{\mathbf{k}} \times \sqrt{\varepsilon}\mathbf{E}_\infty$. A real $\mathbf{E}_\infty$ puts the focus at $\mathbf{r} = 0$.

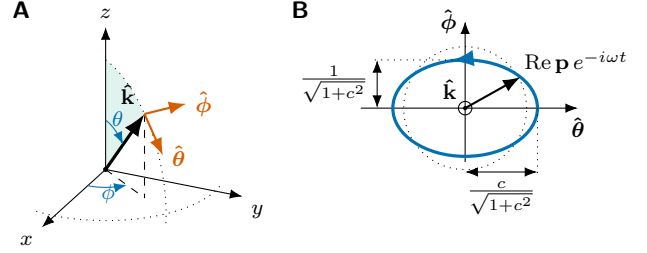


FIG. 1. **A** Angles and unit vectors of Eqs. (3)–(5). A plane wave travels along $\hat{\mathbf{k}}$; the meridional plane (shaded) holds z and $\hat{\mathbf{k}}$. **B** The unit polarization vector $\mathbf{p}$ of one plane wave, Eq. (5), for $\sigma = +1$, viewed with $\hat{\mathbf{k}}$ out of the page. In time, $\text{Re } \mathbf{p} e^{-i\omega t}$ traces an ellipse (blue) with the semi-axes $c/\sqrt{1+c^2}$ along $\hat{\theta}$ and $1/\sqrt{1+c^2}$ along $\hat{\phi}$, here $c = 1.5$. For $c = 1$ it is the dotted circle.

A non-paraxial Gaussian beam is a choice of $\mathbf{E}_\infty$. The choice has two parts, an envelope $a(\theta)$ and a unit polarization vector $\mathbf{p}$:

$$\begin{aligned} \sqrt{\varepsilon}\mathbf{E}_\infty &= \mathcal{A} a(\theta) \mathbf{p}(\theta, \phi), \\ \sqrt{\mu}\mathbf{H}_\infty &= \mathcal{A} a(\theta) \hat{\mathbf{k}} \times \mathbf{p}(\theta, \phi), \end{aligned} \quad (4)$$

with $a(0) = 1$, $|\mathbf{p}| = 1$ and $\mathbf{p}(0, \phi) = \hat{\mathbf{e}}_\sigma$, so that the plane wave along z has the polarization of the paraxial beam (1). Every rule in this note has the form

$$\mathbf{p} = e^{i\sigma\phi} \frac{c(\theta) \hat{\theta} + i\sigma \hat{\phi}}{\sqrt{1+c^2(\theta)}}, \quad (5)$$

where $\hat{\theta} = \cos\phi \cos\theta \hat{\mathbf{x}} + \sin\phi \cos\theta \hat{\mathbf{y}} - \sin\theta \hat{\mathbf{z}}$ and $\hat{\phi} = -\sin\phi \hat{\mathbf{x}} + \cos\phi \hat{\mathbf{y}}$ are the meridional and azimuthal unit vectors, $\hat{\theta} \times \hat{\phi} = \hat{\mathbf{k}}$. The phase $e^{i\sigma\phi}$ gives $\mathbf{p}(0, \phi) = \hat{\mathbf{e}}_\sigma$ for every ϕ . The real weight $c(\theta)$, with $c(0) = 1$, is the ratio of the meridional and the azimuthal components of $\mathbf{p}$ (Fig. 1B). The integral over ϕ in Eq. (3) gives Bessel functions, which leaves one integral over θ for numerics [2]. In the cylindrical coordinates (ρ, φ, z) of $\mathbf{r}$,

$$\begin{aligned} \sqrt{\varepsilon}\mathbf{E}(\mathbf{r}) &= 2\pi\mathcal{A} \int_0^{\theta_{\max}} d\theta \frac{a \sin\theta e^{ikz \cos\theta}}{\sqrt{1+c^2}} \left[\frac{c \cos\theta + 1}{\sqrt{2}} J_0 \hat{\mathbf{e}}_\sigma \right. \\ &\quad \left. - \frac{c \cos\theta - 1}{\sqrt{2}} J_2 e^{2i\sigma\varphi} \hat{\mathbf{e}}_{-\sigma} - ic \sin\theta J_1 e^{i\sigma\varphi} \hat{\mathbf{z}} \right], \end{aligned} \quad (6)$$

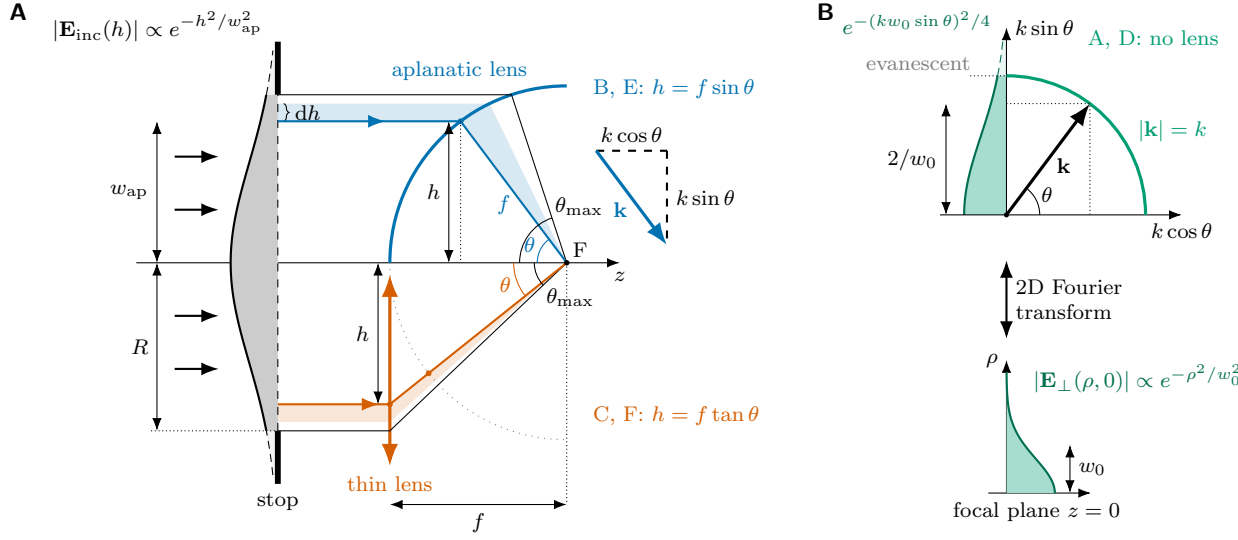


FIG. 2. Where the Gaussian of each beam in Table I lies, for $kw_0 = 2.5$. **A** Beams B, C, E, F. The input field (12) is bent towards the focus F by an aplanatic lens, $h = f \sin \theta$ (upper half), or by a thin lens, $h = f \tan \theta$ (lower half). The drawn rays are at $h = w_{\text{ap}}$, which is the $1/e$ angle θ_b of beams E and F. The shaded tubes carry the power of the same strip dh , which gives the prefactor in Eq. (13); E and F omit it. A stop of radius R blocks $|h| > R$ and sets θ_{max} , with $\text{NA} = n \sin \theta_{\text{max}}$. **B** Beams A, D. The focal-plane field $e^{-\rho^2/w_0^2}$ (bottom) has the spectrum $e^{-(kw_0 \sin \theta)^2/4}$ over the transverse wavevector $k \sin \theta$ (top), the same function of θ as for the aplanatic lens. The part $k \sin \theta > k$ is evanescent and not in Eq. (3).

TABLE I. Six non-paraxial Gaussian beams with the same paraxial waist w_0 . Each beam is Eq. (3) with Eqs. (4) and (5). For B and C an optional stop sets θ_{max} ; it does not change a or c . For E and F the NA sets the $1/e$ angle $\theta_b = \arcsin(\text{NA}/n)$, not a stop.

label	beam	parameters	envelope $a(\theta)$	$c(\theta)$
A	Gaussian spectrum	w_0	$\sqrt{(1 + \cos^2 \theta)/2} e^{-(kw_0 \sin \theta)^2/4}$	$1/\cos \theta$
B	aplanatic lens	w_0 (+ NA of a stop)	$\cos^{1/2} \theta e^{-(kw_0 \sin \theta)^2/4}$	1
C	thin lens	w_0 (+ NA of a stop)	$\cos^{-3/2} \theta e^{-(kw_0 \tan \theta)^2/4}$	1
D	COMSOL plane wave expansion	w_0	$\cos \theta \sqrt{(1 + \cos^2 \theta)/2} e^{-(kw_0 \sin \theta)^2/4}$	$\cos \theta$
E	OTT <code>sintheta</code>	NA	$e^{-(kw_0 \sin \theta)^2/4}$, $kw_0 = 2n/\text{NA}$	1
F	OTT <code>tantheta</code> (default)	NA	$e^{-(kw_0 \tan \theta)^2/4}$, $kw_0 = 2/\tan \theta_b$	1

$$\begin{aligned}
 \sqrt{\mu} \mathbf{H}(\mathbf{r}) = & -2\pi i \sigma \mathcal{A} \int_0^{\theta_{\text{max}}} d\theta \frac{a \sin \theta e^{ikz \cos \theta}}{\sqrt{1+c^2}} \\
 & \times \left[\frac{\cos \theta + c}{\sqrt{2}} J_0 \hat{\mathbf{e}}_\sigma \right. \\
 & \left. - \frac{\cos \theta - c}{\sqrt{2}} J_2 e^{2i\sigma\varphi} \hat{\mathbf{e}}_{-\sigma} - i \sin \theta J_1 e^{i\sigma\varphi} \hat{\mathbf{z}} \right], \quad (7)
 \end{aligned}$$

where $J_m = J_m(k\rho \sin \theta)$ and $\hat{\mathbf{e}}_{-\sigma} = \hat{\mathbf{e}}_\sigma^*$. For $c = 1$, $\sqrt{\mu} \mathbf{H} = -i\sigma\sqrt{\varepsilon} \mathbf{E}$, and the beam is a helicity eigenmode with the eigenvalue σ [5]. For $c \neq 1$ it contains both helicities. Every term has the azimuthal dependence required of an eigenmode of $\hat{J}_z$, the z component of the total angular momentum, with the eigenvalue σ . On the axis only J_0 is not zero, and the field is along $\hat{\mathbf{e}}_\sigma$.

The amplitude $\mathcal{A}$ fixes the power of the beam,

$$P = \int_{x,y} \frac{1}{2} \text{Re}(\mathbf{E} \times \mathbf{H}^*) \cdot d\mathbf{S} = \frac{4\pi^3 |\mathcal{A}|^2}{k^2 \sqrt{\varepsilon\mu}} \int_0^{\theta_{\text{max}}} a^2(\theta) \sin \theta d\theta. \quad (8)$$

In the paraxial limit the focal amplitude of Eq. (1) is $\mathcal{A}_0 = 4\pi\mathcal{A}/(kw_0)^2$.

Every input polarization is $\hat{\mathbf{e}} = \alpha \hat{\mathbf{e}}_+ + \beta \hat{\mathbf{e}}_-$ with $|\alpha|^2 + |\beta|^2 = 1$. Maxwell's equations are linear, so this input gives

$$\mathbf{E} = \alpha \mathbf{E}^{(+)} + \beta \mathbf{E}^{(-)}, \quad \mathbf{H} = \alpha \mathbf{H}^{(+)} + \beta \mathbf{H}^{(-)}, \quad (9)$$

where $\mathbf{E}^{(\sigma)}$, $\mathbf{H}^{(\sigma)}$ are the fields of the beam with the handedness σ and the same a and c . One complex number m fixes the input, $\hat{\mathbf{e}} = (\hat{\mathbf{x}} + m \hat{\mathbf{y}})/\sqrt{1+|m|^2}$, with

$$\alpha = \frac{1 - im}{\sqrt{2(1+|m|^2)}}, \quad \beta = \frac{1 + im}{\sqrt{2(1+|m|^2)}}. \quad (10)$$

III. WHERE THE GAUSSIAN LIES

Many pairs $a(\theta)$, $c(\theta)$ reduce to Eq. (1) for $kw_0 \gg 2$. Each physical construction picks one of them. Table I lists six, and Fig. 2 shows where the Gaussian lies.

No lens (A, D). The Gaussian lies in the angular spectrum: the plane-wave amplitudes are $e^{-(kw_0 \sin \theta)^2/4}$, the Fourier transform of the focal-plane field $e^{-\rho^2/w_0^2}$ (Fig. 2B). The factor $\cos \theta$ of $d\Omega$ cancels, so for beam A the transverse field in the focal plane is exactly Gaussian and along $\hat{\mathbf{e}}_\sigma$; this needs $c = 1/\cos \theta$. The envelope of Table I then follows from $|\mathbf{p}| = 1$. Beam D is the *plane wave expansion* of COMSOL [6, 7]. It projects the input polarization onto the plane normal to $\hat{\mathbf{k}}$,

$$\sqrt{\varepsilon} \mathbf{E}_\infty = \mathcal{A} \cos \theta e^{-(kw_0 \sin \theta)^2/4} [\hat{\mathbf{e}}_\sigma - (\hat{\mathbf{e}}_\sigma \cdot \hat{\mathbf{k}}) \hat{\mathbf{k}}], \quad (11)$$

which is $c = \cos \theta$. COMSOL sums a finite set of plane waves, and it uses $e^{+j\omega t}$, so $i \rightarrow -j$.

Lenses (B, C). The Gaussian lies in the pupil, where the collimated input beam is [1, Eq. (3.52)]

$$\mathbf{E}_{\text{inc}}(h, \phi) = E_0 \hat{\mathbf{e}}_\sigma e^{-h^2/w_{\text{ap}}^2}, \quad w_{\text{ap}} = \frac{2f}{kw_0}. \quad (12)$$

Here h is the distance from the axis, f is the focal length, and w_{ap} is the input radius (called w_0 in Ref. [1]), which the lens focuses to the waist w_0 . The lens sends the ray at the height h into the direction θ , with $h = f \sin \theta$ for an aplanatic lens and $h = f \tan \theta$ for a thin lens (Fig. 2A).

It turns the radial component of $\mathbf{E}_{\text{inc}}$ into the meridional one and keeps the azimuthal one, so $c = 1$ [1, Eq. (3.39)]. A lossless lens keeps the power of each ray: the input power $\frac{1}{2} \sqrt{\varepsilon_1/\mu_1} |\mathbf{E}_{\text{inc}}|^2 h dh d\phi$, with ε_1, μ_1 of the input medium, equals the power in $d\Omega$ in Eq. (8). This gives

$$a(\theta) = \sqrt{\frac{h h'}{f^2 \sin \theta}} e^{-h^2(\theta)/w_{\text{ap}}^2}, \quad \mathcal{A} = \frac{kf}{2\pi} \left(\frac{\varepsilon \varepsilon_1 \mu}{\mu_1} \right)^{1/4} E_0, \quad (13)$$

up to a constant phase, with $h' = dh/d\theta$. The aplanatic lens has $hh' = f^2 \sin \theta \cos \theta$ and the thin lens has $hh' = f^2 \sin \theta / \cos^3 \theta$, which gives rows B and C of Table I. A stop of radius R sets $\theta_{\text{max}} = h^{-1}(R)$. For the aplanatic lens with the filling factor $f_0 = w_{\text{ap}}/R$, this gives $w_0 = \lambda/(\pi f_0 \text{NA})$.

OTT (E, F). The Optical Tweezers Toolbox [8, 9] (class `BscPmGauss`) uses the far field $e^{-h^2(\theta)/h^2(\theta_b)} \mathbf{p}$ with $c = 1$ and the mapping $h = f \sin \theta$ (`sintheta`, E) or $h = f \tan \theta$ (`tantheta`, F). It keeps only the exponential of Eq. (13), without the prefactor. Its NA sets the $1/e$ angle $\theta_b = \arcsin(\text{NA}/n)$, not a stop; the option `truncation_angle` adds a stop. OTT puts the incoming far field near $\theta = \pi$, so compare with Eq. (3) after $\theta \rightarrow \pi - \theta$.

For $\theta \ll 1$ all six envelopes reduce to $e^{-(kw_0\theta)^2/4}$ and all rules to $c = 1$, so all six beams coincide with Eq. (1) up to corrections of order $(kw_0)^{-2}$. The choice of beam is a statement about the optical system, not about Maxwell's equations.

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